Abhijeet Melkani

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Home Page
Github

Education

M.S. & Ph.D. in Physics Sep 2019 → Sep 2024
University of Oregon, Eugene, USA
Thesis: Non-Hermitian Structures in Soft Matter [Advisor: Jayson Paulose]

B. Tech. (with Honours) in Engineering Physics Jul 2015 → May 2019
Indian Institute of Technology Bombay, Mumbai, India

Research Interests

My research lies at the interface of active matter, biophysics, and dynamical metamaterials. I combine analytical approaches—including hydrodynamics, kinetic theory, and tools from non-Hermitian and topological dynamics—with data-driven techniques such as machine learning and agent-based simulations.

Publications & Preprints

- *Shape fluctuations reveal the hydrodynamics of wild fish schools*
  A. Melkani*, A. Delory*, M. Munari, G. Calvet, J. Belusso, F. Gu, D. Bartolo (in preparation)
- *Space-time Floquet operator: Nonreciprocity and fractional topology of space-time crystals*
  A. Melkani, J. Paulose, Phys. Rev. Research 8, L042009 (2026)
- *Space-time symmetry and nonreciprocal parametric resonance in mechanical systems*
  A. Melkani, J. Paulose, Phys. Rev. E 110, 015003 (2024) [Editors' suggestion & 2026 Irwin Oppenheim Award]
- *Degeneracies and symmetry breaking in pseudo-Hermitian matrices*
  A. Melkani, Phys. Rev. Research 5, 023035 (2023)
- *Delocalization of interacting directed polymers on a periodic substrate: Localization length and critical exponents from non-Hermitian spectra*
  A. Melkani*, A. Patapoff*, J. Paulose, Phys. Rev. E 107, 014501 (2023)
- *Electron vortex beams in nonuniform magnetic fields*
  A. Melkani, S. J. van Enk, Phys. Rev. Research 3, 033060 (2021)
- *Eigenstate extraction with neural-network tomography*
  A. Melkani, C. Gneiting, F. Nori, Phys. Rev. A 102, 022412 (2020) [Editors' suggestion]
  *Joint first authors

Scholarships & Awards

American Physical Society Irwin Oppenheim Award	2026
UO Division of Global Engagement Endowed Scholarship	2023
RIKEN Visiting Researcher Award and Travel Award	2018
Outstanding Engineering Student Award sponsored by Uttarakhand Technical University	2016
Ranked 311 in the Indian National Physics Olympiad (Certificate of Excellence)	2015
Ranked 720 in the Indian National Chemistry Olympiad (Certificate of Excellence)	2015
Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship (National Rank 368)	2015